# cost_change_functions_documentation

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17 June, 2025

# R Markdown

Documentation for R script: cost_change_functions.r which contains various functions used in calculating and summarizing the impacts for the non-health pillar outcomes. Everything in here are functions that are used in “cost_change_source_cluster.r”.

The first function here, simulate.impact, is the core of the entire operation. It runs monte carlo simulations for env/cost/labor impact calculations and outputs the resulting distributions. The inputs are going to be specific to the population and dietary factor of interest. In “cost_change_source_cluster.r”, this function is being called for each pair of subgroup/dietary factor.

Meaning of 21 inputs should be mostly self explanatory, but 2 things may need explanation. 1. “substitution.impact.factor.means”, “substitution.impact.factor.ses”, “substitution_unit” refer to substitution impact. While we don’t use this right now (all these inputs are zero), we can calculate substitution effects as well. Say, if we change the diet for just one diet factor, we need replace the calories from other foods. So, we need to calculate the impact of the increase/decrease that comes from the substitution (probably using the impact per unit of the average diet as substition impact factor). Since we keep energy constant for the five alternate scenarios, we don’t actually use this for the 4 pillars paper. But if you’re not keeping energy constant when going from current to counterfactual intake (e.g just changing intake for one dietary factor), you’d want to use substitution effects (and by “not use”, I mean setting all substitution impact factors to zero, not commenting out code related to substitution factors). 2. “current_foodwaste_p”, “current_foodwaste_p_se,”counterfactual_foodwaste_p”, “counterfactual_foodwaste_p_se”: Don’t forget that food waste proportion is conditional on edible. It’s the proportion of the food wasted out of the edible portion. It is NOT food waste / total food including inedible portion.

```
simulate.impact<-function(current.mean, current.se,
                          counterfactual.mean, counterfactual.se=0,
                          impact.factor.names, impact.factor.means,
                          impact.factor.ses, RRunit,
                          substitution.impact.factor.means,
                          substitution.impact.factor.ses, substitution_unit,
                          current_inedible_p, current_inedible_p_se,
                          current_foodwaste_p, current_foodwaste_p_se,
                          counterfactual_inedible_p, counterfactual_inedible_p_se,
                          counterfactual_foodwaste_p, counterfactual_foodwaste_p_se,
                          n.sims,
                          population.sims ##<- must be a vector, so make sure to convert to vector
)
{
```

First simulate intake means for current and counterfactual scenarios. We are creating a matrix of sims because we are going to use them to do calculations for all outcomes from three pillars (with each row corresponding to an outcome of interest).

```
# current.mean.sims<-rnorm(n=n.sims,
#                          mean=current.mean,
#                          sd=current.se)
# counterfactual.mean.sims<-rnorm(n=n.sims,
#                                 mean=counterfactual.mean,
#                                 sd=counterfactual.se)

current.mean.sims<-rnorm(n=n.sims,
                         mean=current.mean,
                         sd=current.se)
current.mean.sims.matrix<-t(matrix(rep(current.mean.sims, times=length(impact.factor.names)),
                           ncol=length(impact.factor.names)))
counterfactual.mean.sims<-rnorm(n=n.sims,
                                mean=counterfactual.mean,
                                sd=counterfactual.se)
counterfactual.mean.sims.matrix<-t(matrix(rep(counterfactual.mean.sims, times=length(impact.fact
                                   ncol=length(impact.factor.names)))
```

Calculate correction factors (from inedible and waste proportion) to go from current intake (our input) to edible amount and total produced (edible + inedible), which is what we need to use to calculate impact on these three pillars.

```
#current.correction<-1/((1-rnorm(n.sims, mean=current_inedible_p, sd=current_inedible_p_se))*(1-
#counterfactual.correction<-1/((1-rnorm(n.sims, mean=counterfactual_inedible_p, sd=counterfactua

current.correction<-1/((1-rnorm(length(impact.factor.names), mean=current_inedible_p, sd=current
counterfactual.correction<-1/((1-rnorm(length(impact.factor.names), mean=counterfactual_inedible

# UNCOMMENT LATER
# ##this is a correction factor to get total edible. "current.correction" above is for getting t
 current.correction.for.waste<-1/(1-rnorm(length(impact.factor.names), mean=current_foodwaste_p,
 counterfactual.correction.for.waste<-1/(1-rnorm(length(impact.factor.names), mean=counterfactua
```

Apply corrections, and based on that, calculate unconsumed, edible, inedible and wasted amounts.

```
#current.mean.sims.after.correction<-current.mean.sims*current.correction
#counterfactual.mean.sims.after.correction<-counterfactual.mean.sims*counterfactual.correction

current.mean.sims.matrix.after.correction<-current.mean.sims.matrix*current.correction
counterfactual.mean.sims.matrix.after.correction<-counterfactual.mean.sims.matrix*counterfactual

current.mean.sims.matrix.unconsumed<-current.mean.sims.matrix.after.correction-current.mean.sims
counterfactual.mean.sims.matrix.unconsumed<-counterfactual.mean.sims.matrix.after.correction-cou

## UNCOMMENT LATER
 current.mean.sims.matrix.edible<-current.mean.sims.matrix*current.correction.for.waste
 counterfactual.mean.sims.matrix.edible<-counterfactual.mean.sims.matrix*counterfactual.correcti

 current.mean.sims.matrix.inedible <- current.mean.sims.matrix.after.correction - current.mean.s
 counterfactual.mean.sims.matrix.inedible <- counterfactual.mean.sims.matrix.after.correction -

 current.mean.sims.matrix.wasted <- current.mean.sims.matrix.edible - current.mean.sims.matrix
 counterfactual.mean.sims.matrix.wasted <- counterfactual.mean.sims.matrix.edible - counterfactu
```

Calculate deltas (difference between current and counterfactual)

```
#delta.sims<-counterfactual.mean.sims-current.mean.sims
 #delta.sims<-counterfactual.mean.sims.after.correction-current.mean.sims.after.correction

 delta.sims.matrix<-counterfactual.mean.sims.matrix.after.correction-current.mean.sims.matrix.aft

 delta.sims.matrix.consumed<-counterfactual.mean.sims.matrix-current.mean.sims.matrix
 delta.sims.matrix.unconsumed<-counterfactual.mean.sims.matrix.unconsumed-current.mean.sims.matri
 delta.sims.matrix.total<-delta.sims.matrix.consumed+delta.sims.matrix.unconsumed

 # UNCOMMENT LATER
  delta.sims.matrix.inedible<-counterfactual.mean.sims.matrix.inedible-current.mean.sims.matrix.i
  delta.sims.matrix.wasted<-counterfactual.mean.sims.matrix.wasted-current.mean.sims.matrix.waste
```

Simulate impact factors.

```
#population.sims<-rnorm(n=n.sims, mean=population, sd=population.se)

impact.factor.sims<-matrix(rnorm(n=n.sims*length(impact.factor.names),
                                 mean=as.numeric(impact.factor.means),
                                 sd=as.numeric(impact.factor.ses)),
                           nrow=length(impact.factor.names))

substitution.impact.factor.sims<-matrix(rnorm(n=n.sims*length(impact.factor.names),
                                              mean=as.numeric(substitution.impact.factor.means),
                                              sd=as.numeric(substitution.impact.factor.ses)),
                                        nrow=length(impact.factor.names))
```

Calculate impact and assign row names to the output matrices (for consumed, unconsumed, total, inedible, and wasted)

```
#impact.sims <- t(t(impact.factor.sims)*delta.sims*t(population.sims))/as.numeric(RRunit)

 #impact.sims <- impact.factor.sims*delta.sims.matrix*population.sims/as.numeric(RRunit)

 impact.sims.consumed <- impact.factor.sims*delta.sims.matrix.consumed*population.sims/as.numeric
 impact.sims.unconsumed <- impact.factor.sims*delta.sims.matrix.unconsumed*population.sims/as.num
 impact.sims.total<-impact.sims.consumed+impact.sims.unconsumed

 # UNCOMMENT LATER
  impact.sims.inedible <- impact.factor.sims*delta.sims.matrix.inedible*population.sims/as.numeri
  impact.sims.wasted <- impact.factor.sims*delta.sims.matrix.wasted*population.sims/as.numeric(RR


 #rownames(impact.sims)<-impact.factor.names
 rownames(impact.sims.consumed)<-impact.factor.names
 rownames(impact.sims.unconsumed)<-impact.factor.names
 rownames(impact.sims.total)<-impact.factor.names

 # UNCOMMENT LATER
  rownames(impact.sims.inedible)<-impact.factor.names
  rownames(impact.sims.wasted)<-impact.factor.names
```

Calculate substitution impact and assign row names to the output matrices

```
#substitution.sims<-t(t(substitution.impact.factor.sims)*(-delta.sims)*t(population.sims))/substi

 #substitution.sims<-substitution.impact.factor.sims*-delta.sims.matrix*population.sims/substitut

 substitution.sims.consumed<-substitution.impact.factor.sims*-delta.sims.matrix.consumed*populati
 substitution.sims.unconsumed<-substitution.impact.factor.sims*-delta.sims.matrix.unconsumed*popu
 substitution.sims.total<-substitution.sims.consumed+substitution.sims.unconsumed

 # UNCOMMENT LATER
  substitution.sims.inedible<-substitution.impact.factor.sims*-delta.sims.matrix.inedible*populat
  substitution.sims.wasted<-substitution.impact.factor.sims*-delta.sims.matrix.wasted*population.

 #rownames(substitution.sims)<-impact.factor.names
 rownames(substitution.sims.consumed)<-impact.factor.names
 rownames(substitution.sims.unconsumed)<-impact.factor.names
 rownames(substitution.sims.total)<-impact.factor.names

 # UNCOMMENT LATER
  rownames(substitution.sims.inedible)<-impact.factor.names
  rownames(substitution.sims.wasted)<-impact.factor.names
```

Combine impact and substitution impact to get total impact

```
#combined.impact.sims<-impact.sims+substitution.sims

combined.impact.sims.consumed<-impact.sims.consumed+substitution.sims.consumed
combined.impact.sims.unconsumed<-impact.sims.unconsumed+substitution.sims.unconsumed
combined.impact.sims.total<-combined.impact.sims.consumed+combined.impact.sims.unconsumed

# UNCOMMENT LATER
 combined.impact.sims.inedible<-impact.sims.consumed+substitution.sims.inedible
 combined.impact.sims.wasted<-impact.sims.unconsumed+substitution.sims.wasted
```

Calculate impact of current and counterfactual intake on the outcomes (as opposed to the the impact of the shift from current to counterfactual, which is what we've been doing up to this point).

```
#combined.impact.sims<-impact.sims+substitution.sims

combined.impact.sims.consumed<-impact.sims.consumed+substitution.sims.consumed
combined.impact.sims.unconsumed<-impact.sims.unconsumed+substitution.sims.unconsumed
combined.impact.sims.total<-combined.impact.sims.consumed+combined.impact.sims.unconsumed

# UNCOMMENT LATER
 combined.impact.sims.inedible<-impact.sims.consumed+substitution.sims.inedible
 combined.impact.sims.wasted<-impact.sims.unconsumed+substitution.sims.wasted

###total impact/cost, not just difference between current and counterfactual
current.impact.sims<-impact.factor.sims*current.mean.sims.matrix.after.correction*population.sim
counterfactual.impact.sims<-impact.factor.sims*counterfactual.mean.sims.matrix.after.correction*
rownames(current.impact.sims)<-impact.factor.names
rownames(counterfactual.impact.sims)<-impact.factor.names

##consumed
current.impact.sims.consumed<-impact.factor.sims*current.mean.sims.matrix*population.sims/as.num
counterfactual.impact.sims.consumed<-impact.factor.sims*counterfactual.mean.sims.matrix*populati
rownames(current.impact.sims.consumed)<-impact.factor.names
rownames(counterfactual.impact.sims.consumed)<-impact.factor.names

##unconsumed
current.impact.sims.unconsumed<-impact.factor.sims*current.mean.sims.matrix.unconsumed*populatio
counterfactual.impact.sims.unconsumed<-impact.factor.sims*counterfactual.mean.sims.matrix.uncons
rownames(current.impact.sims.unconsumed)<-impact.factor.names
rownames(counterfactual.impact.sims.unconsumed)<-impact.factor.names

# UNCOMMENT LATER
## inedible
current.impact.sims.inedible<-impact.factor.sims*current.mean.sims.matrix.inedible*population.si
counterfactual.impact.sims.inedible<-impact.factor.sims*counterfactual.mean.sims.matrix.inedible
rownames(current.impact.sims.inedible)<-impact.factor.names
rownames(counterfactual.impact.sims.inedible)<-impact.factor.names
## wasted
current.impact.sims.wasted<-impact.factor.sims*current.mean.sims.matrix.wasted*population.sims/a
counterfactual.impact.sims.wasted<-impact.factor.sims*counterfactual.mean.sims.matrix.wasted*pop
rownames(current.impact.sims.wasted)<-impact.factor.names
rownames(counterfactual.impact.sims.wasted)<-impact.factor.names
```

Calculating total deltas at population levels by multiplying our mean deltas with population counts

```
# population.sims.output<-as.matrix(population.sims)
# if(dim(population.sims.output)[1]>1)
# {
#   population.sims.output<-population.sims.output[rep(1, times=length(impact.factor.names)),] #
#   rownames(population.sims.output)<-impact.factor.names
# }
# if(dim(population.sims.output)[1]==1)
# {
#   rownames(population.sims.output)<-impact.factor.names
# }
#
#
# delta.sims.output<-as.matrix(delta.sims.matrix*population.sims)
# if(dim(delta.sims.output)[1]>1)
# {
#   delta.sims.output<-delta.sims.output[rep(1, times=length(impact.factor.names)),] ##repeat th
#   rownames(delta.sims.output)<-impact.factor.names
# }
# if(dim(delta.sims.output)[1]==1)
# {
#   rownames(delta.sims.output)<-impact.factor.names
# }

##Now population.sims.output and detla.sims.output are already matrices, don't need above code t
##later, can decide if we want to keep pure deltas, or the modified deltas we use now

population.sims.output<-as.matrix(population.sims)
rownames(population.sims.output)<-impact.factor.names

#delta.sims.output<-as.matrix(delta.sims.matrix*population.sims)
#rownames(delta.sims.output)<-impact.factor.names

delta.sims.output.consumed<-as.matrix(delta.sims.matrix.consumed*population.sims)
rownames(delta.sims.output.consumed)<-impact.factor.names
delta.sims.output.unconsumed<-as.matrix(delta.sims.matrix.unconsumed*population.sims)
rownames(delta.sims.output.unconsumed)<-impact.factor.names
delta.sims.output.total<-as.matrix(delta.sims.matrix.total*population.sims)
rownames(delta.sims.output.total)<-impact.factor.names

# UNCOMMENT LATER
delta.sims.output.inedible<-as.matrix(delta.sims.matrix.inedible*population.sims)
rownames(delta.sims.output.inedible)<-impact.factor.names
delta.sims.output.wasted<-as.matrix(delta.sims.matrix.wasted*population.sims)
rownames(delta.sims.output.wasted)<-impact.factor.names
```

Take all output created and put it into a list of 31 matrices, each corresponding to simulations for a specific output of interest for the particular dietary factor - counterfactual diet pair, and return that list. Let's spell out what all the outcomes are here:

1. impact.sims.total = impact of change from current to counterfactual for dietary factor of interest

2. substitution.sims.total = impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest
3. combined.impact.sims.total = combined impact of dietary factor shift and corresponding substitution impact (impact.sim.total+substitution.sims.total)
4. population.sims.total = population count
5. delta.sims.total = difference in dietary factor amount produced for current vs counterfactual
6. impact.sims.consumed = impact of change from current to counterfactual for dietary factor of interest, but just for the consumed part of the dietary factor
7. substitution.sims.consumed = impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the consumed part of the dietary factor
8. combined.impact.sims.consumed = combined impact of dietary factor shift and corresponding substitution impact (impact.sim.consumed+substitution.sims.consumed), but just for the consumed part of the dietary factor
9. delta.sims.consumed = difference in dietary factor amount produced for current vs counterfactual, but just for the consumed part of the dietary factor
10. impact.sims.unconsumed = impact of change from current to counterfactual for dietary factor of interest, but just for the unconsumed part of the dietary factor
11. substitution.sims.consumed = impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the consumed part of the dietary factor
12. combined.impact.sims.consumed = combined impact of dietary factor shift and corresponding substitution impact (impact.sim.unconsumed+substitution.sims.unconsumed), but just for the unconsumed part of the dietary factor
13. delta.sims.unconsumed = difference in dietary factor amount produced for current vs counterfactual, but just for the unconsumed part of the dietary factor
14. current.intake.impact.total = impact of current intake for dietary factor of interest
15. CF.intake.impact.total = impact of counterfactual intake for dietary factor of interest
16. current.intake.impact.consumed = impact of current intake for dietary factor of interest, but just for the consumed part
17. CF.intake.impact.consumed = impact of counterfactual intake for dietary factor of interest, but just for the consumed part
18. current.intake.impact.unconsumed = impact of current intake for dietary factor of interest, but just for the unconsumed part
19. CF.intake.impact.unconsumed = impact of counterfactual intake for dietary factor of interest, but just for the unconsumed part
20. impact.sims.inedible = impact of change from current to counterfactual for dietary factor of interest, but just for the inedible part of the dietary factor
21. substitution.sims.inedible = impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the inedible part of the dietary factor

22. combined.impact.sims.inedible = combined impact of dietary factor shift and corresponding substitution impact (impact.sim.inedible+substitution.sims.inedible), but just for the inedible part of the dietary factor
23. delta.sims.inedible = difference in dietary factor amount produced for current vs counterfactual, but just for the inedible part of the dietary factor
24. impact.sims.wasted = impact of change from current to counterfactual for dietary factor of interest, but just for the wasted part of the dietary factor
25. substitution.sims.wasted = impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the wasted part of the dietary factor
26. combined.impact.sims.wasted = combined impact of dietary factor shift and corresponding substitution impact (impact.sim.wasted+substitution.sims.wasted), but just for the wated part of the dietary factor
27. delta.sims.wasted = difference in dietary factor amount produced for current vs counterfactual, but just for the inedible part of the dietary factor
28. current.intake.impact.inedible = impact of current intake for dietary factor of interest, but just for the inedible part
29. CF.intake.impact.inedible = impact of counterfactual intake for dietary factor of interest, but just for the inedible part
30. current.intake.impact.wasted = impact of current intake for dietary factor of interest, but just for the wasted part
31. CF.intake.impact.wasted = impact of counterfactual intake for dietary factor of interest, but just for the wasted part

```
  # impact.list<-list("impact.sims" = impact.sims,
  #                   "substitution.sims" = substitution.sims,
  #                   "combined.impact.sims" = combined.impact.sims,
  #                   "population.sims" = population.sims.output,
  #                   "delta.sims" = delta.sims.output)

  impact.list<-list("impact.sims.total" = impact.sims.total,
                    "substitution.sims.total" = substitution.sims.total,
                    "combined.impact.sims.total" = combined.impact.sims.total,
                    "population.sims.total" = population.sims.output,
                    "delta.sims.total" = delta.sims.output.total,

                    "impact.sims.consumed" = impact.sims.consumed,
                    "substitution.sims.consumed" = substitution.sims.consumed,
                    "combined.impact.sims.consumed" = combined.impact.sims.consumed,
                    "delta.sims.consumed" = delta.sims.output.consumed,

                    "impact.sims.unconsumed" = impact.sims.unconsumed,
                    "substitution.sims.unconsumed" = substitution.sims.unconsumed,
                    "combined.impact.sims.unconsumed" = combined.impact.sims.unconsumed,
                    "delta.sims.unconsumed" = delta.sims.output.unconsumed,

                    "current.intake.impact.total" = current.impact.sims,
                    "CF.intake.impact.total" = counterfactual.impact.sims,
                     "current.intake.impact.consumed" = current.impact.sims.consumed,
                     "CF.intake.impact.consumed" = counterfactual.impact.sims.consumed,
                     "current.intake.impact.unconsumed" = current.impact.sims.unconsumed,
                     "CF.intake.impact.unconsumed" = counterfactual.impact.sims.unconsumed,

                    #UNCOMMENT LATER,
                    "impact.sims.inedible" = impact.sims.inedible,
                    "substitution.sims.inedible" = substitution.sims.inedible,
                    "combined.impact.sims.inedible" = combined.impact.sims.inedible,
                    "delta.sims.inedible" = delta.sims.output.inedible,

                    "impact.sims.wasted" = impact.sims.wasted,
                    "substitution.sims.wasted" = substitution.sims.wasted,
                    "combined.impact.sims.wasted" = combined.impact.sims.wasted,
                    "delta.sims.wasted" = delta.sims.output.wasted,

                    "current.intake.impact.inedible" = current.impact.sims.inedible,
                    "CF.intake.impact.inedible" = counterfactual.impact.sims.inedible,
                    "current.intake.impact.wasted" = current.impact.sims.wasted,
                    "CF.intake.impact.wasted" = counterfactual.impact.sims.wasted
                    )

  return(impact.list)
}
```

Next is a function that calculate per capita sims by dividing impact sims with population count.

```
get.percapita.sims<-function(impact.sims, population.sims, pop.sims.names, new.pop.sims.names, sim
{
  names(population.sims)[names(population.sims) %in% pop.sims.names]<-new.pop.sims.names ##replace
  impact.sims.plus.pop<-merge(impact.sims, population.sims)

  impact.sims.percapita<-impact.sims.plus.pop
  impact.sims.percapita[,sims.names]<-impact.sims.percapita[,sims.names]/impact.sims.percapita[,ne
  impact.sims.percapita<-impact.sims.percapita[,!(names(impact.sims.percapita) %in% new.pop.sims.n

  return(impact.sims.percapita)
}
```

Next, a function that calculates summary stats from simulation outputs, taking in simulation results for a specific diet factor as an input. Returns a list of 22 data tables, each corresponding to a different outcome (deltas, impact, substitution impact, combined impact, etc…). The 22 data tables are spelled out below:

1. delta.sims.summary = summary stats for change in amount produced for dietary factor of interest
2. impact.sims.summary = summary stats for impact of change from current to counterfactual for dietary factor of interest
3. substitution.impact.sims.summary = summary stats for impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest
4. combined.impact.sims.summary = summary stats for combined impact of dietary factor shift and corresponding substitution impact
5. impact.sims.consumed.summary = summary stats for impact of change from current to counterfactual for dietary factor of interest, but just for the consumed part of the dietary factor
6. substitution.impact.sims.consumed.summary = summary stats for combined impact of dietary factor shift and corresponding substitution impact, but just for the consumed part of the dietary factor
7. combined.impact.sims.consumed.summary = summary stats for combined impact of dietary factor shift and corresponding substitution impact, but just for the consumed part of the dietary factor
8. impact.sims.unconsumed.summary = summary stats for impact of change from current to counterfactual for dietary factor of interest, but just for the unconsumed part of the dietary factor
9. substitution.impact.sims.unconsumed.summary = summary stats for impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the unconsumed part of the dietary factor
10. combined.impact.sims.unconsumed.summary = summary stats for combined impact of dietary factor shift and corresponding substitution impact, but just for the unconsumed part of the dietary factor
11. current.intake.impact.sims.summary = summary stats for impact of current intake for dietary factor of interest
12. current.intake.impact.consumed.sims.summary = summary stats for impact of current intake for dietary factor of interest, but just for the consumed part of the dietary factor
13. current.intake.impact.unconsumed.sims.summary = summary stats for impact of current intake for dietary factor of interest, but just for the unconsumed part of the dietary factor
14. CF.intake.impact.sims.summary = summary stats for impact of counterfactual intake for dietary factor of interest
15. CF.intake.impact.consumed.sims.summary = summary stats for impact of counterfactual intake for dietary factor of interest, but just for the consumed part of the dietary factor
16. CF.intake.impact.unconsumed.sims.summary = summary stats for impact of counterfactual intake for dietary factor of interest, but just for the unconsumed part of the dietary factor

17. impact.sims.inedible.summary = summary stats for impact of change from current to counterfactual for dietary factor of interest, but just for the inedible part of the dietary factor
18. substitution.impact.sims.inedible.summary = summary stats for impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the inedible part of the dietary factor
19. combined.impact.sims.inedible.summary = summary stats for combined impact of dietary factor shift and corresponding substitution impact, but just for the inedible part of the dietary factor
20. impact.sims.wasted.summary = summary stats for impact of change from current to counterfactual for dietary factor of interest, but just for the wasted part of the dietary factor
21. substitution.impact.sims.wasted.summary = summary stats for impact from substitution of calories needed to maintain energy going from current to counterfactual for dietary factor of interest, but just for the wasted part of the dietary factor
22. combined.impact.sims.wasted.summary = summary stats for combined impact of dietary factor shift and corresponding substitution impact, but just for the wasted part of the dietary factor

```
#let's make a functino out of this, since we're gonna repeat this whole thing for per capita
get.summary.stats<-function(impact.sims, substitution.impact.sims, combined.impact.sims, delta.sim
                            impact.sims.consumed, substitution.impact.sims.consumed, combined.impa
                            impact.sims.unconsumed, substitution.impact.sims.unconsumed, combined.
                            current.intake.impact.sims, current.intake.impact.sims.consumed, curre
                            CF.intake.impact.sims, CF.intake.impact.sims.consumed, CF.intake.impac
                            impact.sims.inedible, substitution.impact.sims.inedible, combined.impa
                            impact.sims.wasted, substitution.impact.sims.wasted, combined.impact.s
                            current.intake.impact.sims.inedible, current.intake.impact.sims.wasted
                            CF.intake.impact.sims.inedible, CF.intake.impact.sims.wasted,
                            vars, convert.to.kcal.units="convert.to.kcal.units")
{
  impact.sims.dt<-as.data.table(impact.sims)
  impact.sims.summary <- impact.sims.dt[, ":="(mean_impact = rowMeans(.SD, na.rm = TRUE),
                                               sd_impact = apply(.SD, 1, sd),
                                               median_impact = apply(.SD, 1, median),
                                               lowerci_95_impact = apply(.SD, 1, function(x) quant
                                               upperci_95_impact = apply(.SD, 1, function(x) quant
                                        #lowerci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                        #upperci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                        .SDcols = vars
  ]
  impact.sims.summary <- impact.sims.summary[, (vars) := NULL]

  substitution.impact.sims.dt<-as.data.table(substitution.impact.sims)
  substitution.impact.sims.summary <- substitution.impact.sims.dt[, ":="(mean_substitution_impact
                                                                          sd_substitution_impact =
                                                                          median_substitution_impac
                                                                          lowerci_95_substitution_i
                                                                          upperci_95_substitution_i
                                                                    #lowerci_90_substitution_impact
                                                                    #upperci_90_substitution_impact
                                                                    .SDcols = vars
  ]
  substitution.impact.sims.summary <- substitution.impact.sims.summary[, (vars) := NULL]

  combined.impact.sims.dt<-as.data.table(combined.impact.sims)
  combined.impact.sims.summary <- combined.impact.sims.dt[, ":="(mean_combined_impact = rowMeans(.
                                                                   sd_combined_impact = apply(.SD, 1
                                                                   median_combined_impact = apply(.S
                                                                   lowerci_95_combined_impact = appl
                                                                   upperci_95_combined_impact = appl
                                                              #lowerci_90_combined_impact = apply(.SD,
                                                              #upperci_90_combined_impact = apply(.SD,
                                                              .SDcols = vars
  ]
  combined.impact.sims.summary <- combined.impact.sims.summary[, (vars) := NULL]

  delta.sims.dt<-as.data.table(delta.sims)
  delta.sims.summary <- delta.sims.dt[, ":="(mean_delta = rowMeans(.SD, na.rm = TRUE),
                                             sd_delta = apply(.SD, 1, sd),
                                             median_delta = apply(.SD, 1, median),
```

```
                                        lowerci_95_delta = apply(.SD, 1, function(x) quantile
                                        upperci_95_delta = apply(.SD, 1, function(x) quantile
                                    #lowerci_90_impact = apply(.SD, 1, function(x) quantile(x, .
                                    #upperci_90_impact = apply(.SD, 1, function(x) quantile(x, .
                                    .SDcols = vars
]
delta.sims.summary <- delta.sims.summary[, (vars) := NULL]

######_consumed
impact.sims.consumed.dt<-as.data.table(impact.sims.consumed)
impact.sims.consumed.summary <- impact.sims.consumed.dt[, ":="(mean_impact_consumed = rowMeans(.
                                        sd_impact_consumed = apply(.SD, 1, sd),
                                        median_impact_consumed = apply(.SD, 1, median),
                                        lowerci_95_impact_consumed = apply(.SD, 1, function
                                        upperci_95_impact_consumed = apply(.SD, 1, function
                                  #lowerci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                  #upperci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                  .SDcols = vars
]
impact.sims.consumed.summary <- impact.sims.consumed.summary[, (vars) := NULL]

substitution.impact.sims.consumed.dt<-as.data.table(substitution.impact.sims.consumed)
substitution.impact.sims.consumed.summary <- substitution.impact.sims.consumed.dt[, ":="(mean_su
                                                                    sd_substitution_impact_co
                                                                    median_substitution_impac
                                                                    lowerci_95_substitution_i
                                                                    upperci_95_substitution_i
                                                                #lowerci_90_substitution_impact
                                                                #upperci_90_substitution_impact
                                                                .SDcols = vars
]
substitution.impact.sims.consumed.summary <- substitution.impact.sims.consumed.summary[, (vars)

combined.impact.sims.consumed.dt<-as.data.table(combined.impact.sims.consumed)
combined.impact.sims.consumed.summary <- combined.impact.sims.consumed.dt[, ":="(mean_combined_i
                                                                    sd_combined_impact_consumed = app
                                                                    median_combined_impact_consumed =
                                                                    lowerci_95_combined_impact_consum
                                                                    upperci_95_combined_impact_consum
                                                                #lowerci_90_combined_impact = apply(.SD,
                                                                #upperci_90_combined_impact = apply(.SD,
                                                                .SDcols = vars
]
combined.impact.sims.consumed.summary <- combined.impact.sims.consumed.summary[, (vars) := NULL]

########unconsumed
impact.sims.unconsumed.dt<-as.data.table(impact.sims.unconsumed)
impact.sims.unconsumed.summary <- impact.sims.unconsumed.dt[, ":="(mean_impact_unconsumed = rowM
                                                                    sd_impact_unconsumed = apply(.SD,
                                                                    median_impact_unconsumed = apply(
                                                                    lowerci_95_impact_unconsumed = ap
                                                                    upperci_95_impact_unconsumed = ap
```

```
                                                    #lowerci_90_impact = apply(.SD, 1, funct
                                                    #upperci_90_impact = apply(.SD, 1, funct
                                                    .SDcols = vars
]
impact.sims.unconsumed.summary <- impact.sims.unconsumed.summary[, (vars) := NULL]

substitution.impact.sims.unconsumed.dt<-as.data.table(substitution.impact.sims.unconsumed)
substitution.impact.sims.unconsumed.summary <- substitution.impact.sims.unconsumed.dt[, ":="(mea
                                                                                    sd_subs
                                                                                    median_
                                                                                    lowerci
                                                                                    upperci
                                                                                #lowerci_90_su
                                                                                #upperci_90_su
                                                                                .SDcols = vars
]
substitution.impact.sims.unconsumed.summary <- substitution.impact.sims.unconsumed.summary[, (va

combined.impact.sims.unconsumed.dt<-as.data.table(combined.impact.sims.unconsumed)
combined.impact.sims.unconsumed.summary <- combined.impact.sims.unconsumed.dt[, ":="(mean_combin
                                                                                sd_combined_imp
                                                                                median_combined
                                                                                lowerci_95_comb
                                                                                upperci_95_comb
                                                                            #lowerci_90_combined_i
                                                                            #upperci_90_combined_i
                                                                            .SDcols = vars
]
combined.impact.sims.unconsumed.summary <- combined.impact.sims.unconsumed.summary[, (vars) := N

##### current.intake

current.intake.impact.sims.dt<-as.data.table(current.intake.impact.sims)
current.intake.impact.sims.summary <- current.intake.impact.sims.dt[, ":="(mean_impact_current_i
                                                                        sd_impact_current_intake
                                                                        median_impact_current_int
                                                                        lowerci_95_impact_current
                                                                        upperci_95_impact_current
                                                                    #lowerci_90_substitution_impact
                                                                    #upperci_90_substitution_impact
                                                                    .SDcols = vars
]
current.intake.impact.sims.summary <- current.intake.impact.sims.summary[, (vars) := NULL]

current.intake.impact.consumed.sims.dt<-as.data.table(current.intake.impact.sims.consumed)
current.intake.impact.consumed.sims.summary <- current.intake.impact.consumed.sims.dt[, ":="(mea
                                                                                        sd_
                                                                                        med
                                                                                        low
                                                                                        upp
                                                                                    #lowerci_9
                                                                                    #upperci_9
```

```
                                                                                .SDcols =
]
current.intake.impact.consumed.sims.summary <- current.intake.impact.consumed.sims.summary[, (va


current.intake.impact.unconsumed.sims.dt<-as.data.table(current.intake.impact.sims.unconsumed)
current.intake.impact.unconsumed.sims.summary <- current.intake.impact.unconsumed.sims.dt[, ":="
                                                                                sd_
                                                                                med
                                                                                low
                                                                                upp
                                                                            #lowerci_9
                                                                            #upperci_9
                                                                            .SDcols =
]
current.intake.impact.unconsumed.sims.summary <- current.intake.impact.unconsumed.sims.summary[,

#####CF.intake. _CF_intake
CF.intake.impact.sims.dt<-as.data.table(CF.intake.impact.sims)
CF.intake.impact.sims.summary <- CF.intake.impact.sims.dt[, ":="(mean_impact_CF_intake = rowMean
                                        sd_impact_CF_intake = apply(.SD, 1, sd),
                                        median_impact_CF_intake = apply(.SD, 1, median),
                                        lowerci_95_impact_CF_intake = apply(.SD, 1, functio
                                        upperci_95_impact_CF_intake = apply(.SD, 1, functio
                                    #lowerci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                    #upperci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                    .SDcols = vars
]
CF.intake.impact.sims.summary <- CF.intake.impact.sims.summary[, (vars) := NULL]

CF.intake.impact.consumed.sims.dt<-as.data.table(CF.intake.impact.sims.consumed)
CF.intake.impact.consumed.sims.summary <- CF.intake.impact.consumed.sims.dt[, ":="(mean_impact_C
                                                                                sd_
                                                                                med
                                                                                low
                                                                                upp
                                                                            #lowerci_9
                                                                            #upperci_9
                                                                            .SDcols =
]
CF.intake.impact.consumed.sims.summary <- CF.intake.impact.consumed.sims.summary[, (vars) := NUL


CF.intake.impact.unconsumed.sims.dt<-as.data.table(CF.intake.impact.sims.unconsumed)
CF.intake.impact.unconsumed.sims.summary <- CF.intake.impact.unconsumed.sims.dt[, ":="(mean_impa




                                                                                #lower
                                                                                #upper
                                                                                .SDcol
```

```
]
CF.intake.impact.unconsumed.sims.summary <- CF.intake.impact.unconsumed.sims.summary[, (vars) :=

####

########inedible
impact.sims.inedible.dt<-as.data.table(impact.sims.inedible)
impact.sims.inedible.summary <- impact.sims.inedible.dt[, ":="(mean_impact_inedible = rowMeans(.
                                                               sd_impact_inedible = apply(.S
                                                               median_impact_inedible = appl
                                                               lowerci_95_impact_inedible =
                                                               upperci_95_impact_inedible =
                                                          #lowerci_90_impact = apply(.SD, 1, f
                                                          #upperci_90_impact = apply(.SD, 1, f
                                                          .SDcols = vars
]
impact.sims.inedible.summary <- impact.sims.inedible.summary[, (vars) := NULL]

substitution.impact.sims.inedible.dt<-as.data.table(substitution.impact.sims.inedible)
substitution.impact.sims.inedible.summary <- substitution.impact.sims.inedible.dt[, ":="(mean_su
                                                                                        sd_
                                                                                        med
                                                                                        low
                                                                                        upp
                                                                                   #lowerci_9
                                                                                   #upperci_9
                                                                                   .SDcols =
]
substitution.impact.sims.inedible.summary <- substitution.impact.sims.inedible.summary[, (vars)

combined.impact.sims.inedible.dt<-as.data.table(combined.impact.sims.inedible)
combined.impact.sims.inedible.summary <- combined.impact.sims.inedible.dt[, ":="(mean_combined_i
                                                                                 sd_combined_impact_
                                                                                 median_combined_imp
                                                                                 lowerci_95_combined
                                                                                 upperci_95_combined
                                                                            #lowerci_90_combined_impac
                                                                            #upperci_90_combined_impac
                                                                            .SDcols = vars
]
combined.impact.sims.inedible.summary <- combined.impact.sims.inedible.summary[, (vars) := NULL]


########wasted
impact.sims.wasted.dt<-as.data.table(impact.sims.wasted)
impact.sims.wasted.summary <- impact.sims.wasted.dt[, ":="(mean_impact_wasted = rowMeans(.SD, na
                                                           sd_impact_wasted = apply(.SD, 1,
                                                           median_impact_wasted = apply(.SD,
                                                           lowerci_95_impact_wasted = apply(
                                                           upperci_95_impact_wasted = apply(
                                                      #lowerci_90_impact = apply(.SD, 1, funct
                                                      #upperci_90_impact = apply(.SD, 1, funct
```

```
                                                    .SDcols = vars
  ]
  impact.sims.wasted.summary <- impact.sims.wasted.summary[, (vars) := NULL]

  substitution.impact.sims.wasted.dt<-as.data.table(substitution.impact.sims.wasted)
  substitution.impact.sims.wasted.summary <- substitution.impact.sims.wasted.dt[, ":="(mean_substi
                                                                                    sd_subs
                                                                                    median_
                                                                                    lowerci
                                                                                    upperci
                                                                                #lowerci_90_su
                                                                                #upperci_90_su
                                                                                .SDcols = vars
  ]
  substitution.impact.sims.wasted.summary <- substitution.impact.sims.wasted.summary[, (vars) := N

  combined.impact.sims.wasted.dt<-as.data.table(combined.impact.sims.wasted)
  combined.impact.sims.wasted.summary <- combined.impact.sims.wasted.dt[, ":="(mean_combined_impac
                                                                              sd_combined
                                                                              median_comb
                                                                              lowerci_95_
                                                                              upperci_95_
                                                                          #lowerci_90_combin
                                                                          #upperci_90_combin
                                                                          .SDcols = vars
  ]
  combined.impact.sims.wasted.summary <- combined.impact.sims.wasted.summary[, (vars) := NULL]


  all.impact.sims.summary<-Reduce(function(...) merge(..., all = T),list(delta.sims.summary,
                                                                         impact.sims.summary, subs
                                                                         impact.sims.consumed.summ
                                                                         impact.sims.unconsumed.su
                                                                         current.intake.impact.sim
                                                                         CF.intake.impact.sims.sum
                                                                         impact.sims.inedible.summ
                                                                         impact.sims.wasted.summar

  return(all.impact.sims.summary)
}
```

Simple version of above function to use on just one set of outputs (as opposed to 22). Inputs are 1. impact.sims: the data frame with output sims of interest, and 2. vars: the variable names for the sims (e.g, V1, V2, …. V1000).

```
get.summary.stats.simple<-function(impact.sims, vars)
{
  impact.sims.dt<-as.data.table(impact.sims)
  impact.sims.summary <- impact.sims.dt[, ":="(mean_impact = rowMeans(.SD, na.rm = TRUE),
                                               sd_impact = apply(.SD, 1, sd),
                                               median_impact = apply(.SD, 1, median),
                                               lowerci_95_impact = apply(.SD, 1, function(x) quant
                                               upperci_95_impact = apply(.SD, 1, function(x) quant
                                        #lowerci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                        #upperci_90_impact = apply(.SD, 1, function(x) quantile(x,
                                        .SDcols = vars
  ]
  impact.sims.summary <- impact.sims.summary[, (vars) := NULL]

  return(impact.sims.summary)
}
```

And finally, a function that sums the output by all strata combinations of interest and outputs summary stats. Note that input “strata.combos.list” is a list of the strata combinations of interest, with each element being a string vector denoting the strata for that combination. “impact.sims.allgroups” is data frame with output sims that you want to summarize.

```
summary.stats.by.strata<-function(impact.sims.allgroups, strata.combos.list, pop.sims)
{
```

Convert sims to data tables, create some string vectors for labeling simulated values, and create lists for storing results.

```
  cols.to.sum<-paste("X", 1:n.sims, sep="")
  pop.cols.to.sum<-paste("pop", 1:n.sims, sep="")

  sims.data.table<-as.data.table(impact.sims.allgroups)
  pop.sims.data.table<-as.data.table(pop.sims)

  strata.combos.sims<-list()
  strata.combos.sims.long<-list()
  strata.combos.summary.stats.long<-list()
  strata.combos.summary.stats.wide<-list()
  strata.combos.summary.stats.misc<-list()


  pop.strata.combos.sims<-list()

  percapita.sims<-list()
  percapita.sims.long<-list()
  percapita.summary.stats.long<-list()
  percapita.summary.stats.wide<-list()
  percapita.summary.stats.misc<-list()
```

Start a for loop to traverse through each strata combination of interest.

```
for(j in 1:length(strata.combos.list))
{
```

Sum by combinations of strata in strata.combos.list[[j]]

```
  labels<-c(strata.combos[[j]], "outcome", "outcome_unit")

  strata.combos.sims[[j]]<-sims.data.table[,lapply(.SD, sum), by=c(strata.combos.list[[j]], "out
                                          .SDcols=cols.to.sum]
```

Sum population sims by combinations of strata in strata.combos.list[[j]]. Note that we can't sum population counts by "Foodgroup" or "broad_foodgroup" since we would be double (or triple, quadruple) counting (these are not population attributes) if we did that, so we remove those from the strata combination when summing population counts. If the strata combination of interest is just the foodgroups, then we sum over all subgroups.

```
  pop.strata<-strata.combos[[j]][!(strata.combos.list[[j]] %in% c("Foodgroup", "broad_foodgroup"

  if(is.null(pop.strata))
  {
    pop.strata.combos.sims[[j]]<-pop.sims.data.table[,lapply(.SD, sum), .SDcols=cols.to.sum]
  }else
  {
    pop.strata.combos.sims[[j]]<-pop.sims.data.table[,lapply(.SD, sum), by=c(pop.strata), .SDcol
  }
  names(pop.strata.combos.sims[[j]])<-gsub(x=names(pop.strata.combos.sims[[j]]), pattern="X", re
```

merge the summed inputs and pop counts into one file.

```
  if(length(intersect(names(strata.combos.sims[[j]]), names(pop.strata.combos.sims[[j]])))==0)
  {
    merged<-cbind(strata.combos.sims[[j]], pop.strata.combos.sims[[j]])
  }
  if(length(intersect(names(strata.combos.sims[[j]]), names(pop.strata.combos.sims[[j]])))!=0)
  {
    merged<-merge(strata.combos.sims[[j]], pop.strata.combos.sims[[j]])
  }
```

Calculate per capita results of sims summed by strata

```
  merged.names<-merged[,..labels]
  percapita.sims[[j]]<-merged[,..cols.to.sum]/merged[,..pop.cols.to.sum]
  percapita.sims[[j]]<-cbind(merged.names, percapita.sims[[j]])
```

Reshape sims into long format, calculate summary stats, and label outcomes appropriately, then reshape back to wide format so that each summary stat has its own column.

```
  strata.combos.sims.long[[j]]<-melt(strata.combos.sims[[j]], id.vars = c(strata.combos[[j]], "out
    strata.combos.summary.stats.long[[j]]<-strata.combos.sims.long[[j]][, .(quantile(value, c(.025
    strata.combos.summary.stats.misc[[j]]<-strata.combos.sims.long[[j]][, .(mean(value), sd(value)
    names(strata.combos.summary.stats.misc[[j]])[(dim(strata.combos.summary.stats.misc[[j]])[2]-1)

    strata.combos.summary.stats.long[[j]]<-cbind(strata.combos.summary.stats.long[[j]],
                                                 as.factor(rep(c("lower_bound (2.5th percentile)",
    #strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], "Comorbid
    strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], paste(past
                                                 value.var="V1")
    #strata.combos.summary.stats.wide[[j]]<-merge(strata.combos.summary.stats.wide[[j]], strata.co
    strata.combos.summary.stats.wide[[j]]<-merge(strata.combos.summary.stats.wide[[j]], strata.com
                                               by=intersect(names(strata.combos.summary.stats.wide[[j
```

Repeat for per capita sims.

```
    percapita.sims.long[[j]]<-melt(percapita.sims[[j]], id.vars = c(strata.combos.list[[j]], "outc
    percapita.summary.stats.long[[j]]<-percapita.sims.long[[j]][, .(quantile(value, c(.025, .5, .9
    percapita.summary.stats.misc[[j]]<-percapita.sims.long[[j]][, .(mean(value), sd(value)), by=c(
    names(percapita.summary.stats.misc[[j]])[(dim(strata.combos.summary.stats.misc[[j]])[2]-1):dim

    percapita.summary.stats.long[[j]]<-cbind(percapita.summary.stats.long[[j]],
                                             as.factor(rep(c("lower_bound (2.5th percentile)", "me
                                                           dim(percapita.summary.stats.long[[j]])[
    #strata.combos.summary.stats.wide[[j]]<-dcast(strata.combos.summary.stats.long[[j]], "Comorbid
    percapita.summary.stats.wide[[j]]<-dcast(percapita.summary.stats.long[[j]], paste(paste(strata
                                             value.var="V1")
    percapita.summary.stats.wide[[j]]<-merge(percapita.summary.stats.wide[[j]], percapita.summary.
                                             by=intersect(names(percapita.summary.stats.wide[[j]])
```

End loop

```
}
```

After looping through all strata combos of interest, we return the result as a 4 element list. 1 = summary output, 2 = full sims, 3 = per capita summary output, 4 = per capita full sims. Each of these 4 elements are themselves lists with with element corresponding to a strata combination of interest.

```
  return(list("summary.output" = strata.combos.summary.stats.wide,
              "full.sims.output" = strata.combos.sims,
              "summary.output.percapita"= percapita.summary.stats.wide,
              "full.sims.output.percapita" = percapita.sims)
  )
}
```